This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

Technical Proposal - Smart Health Checkup Recommendation Dashboard

1. Executive Summary
   - 1.1 Introduction
   - 1.2 Background Information
   - 1.3 To-Be Vision & Core Value Proposition
2. Project Scope & Assumptions
   - 2.1 Scope of Work (In-Scope Deliverables)
   - 2.2 Out of Scope
   - 2.3 Key Assumptions & Dependencies
3. High-Level System Architecture
   - 3.1 Data flow
   - 3.2 Conceptual Architecture
     - C4 Context Diagram
     - C4 Container Diagram
   - 3.3 Core Modules Overview
4. Data Platform & Integration Architecture
   - High-level Data Platform Architecture
   - 4.1 Data Ingestion Strategy (Connecting to Navavej HIS)
   - 4.2 Data Pipeline at a Glance
   - 4.3 Data Conversion & Health Trend Pre-Computing
     - 4.3.1 Clinical Code Mapping (LOINC, SNOMED CT, ICD-10)
     - 4.3.2 Health Trend Pre-Computing (Velocity, Gaps, eGFR slope)
   - 4.4 Dual-Storage Strategy
     - 4.4.1 Core FHIR Database (Immutable Clinical Records)
     - 4.4.2 Trending Store Database (AI Analytical Feature Store)
   - 5. Data Platform Tech Stack
6. Security, Privacy & Data Governance
   - 6.1 SMART on FHIR Authorization (Access Control Gatekeeper)
   - 6.2 Data Masking & Pseudo-Anonymization Pipelines
   - 6.3 Comprehensive Audit Logging
7. AI & Recommendation Engine Architecture
   - 7.1 High-Level AI Workflow
   - 7.2 Model Context Protocol (MCP) Middleware (Secure Context Retrieval)
   - 7.3 Hybrid Recommendation Logic (Rule-based Safety + AI Inference)
   - 7.4 Central AI Gateway
     - 7.4.2 Pluggable LLM Configuration Management
     - 7.4.1 Explainable AI (XAI) & Confidence Scoring Generation
   - 7.5 AI Tech Stack
8. Frontend & Dashboards (User Experience)
   - 8.1 Responsive Web Application Architecture (Tech Stack)
   - 8.2 Healthcare Provider Dashboard (Summaries, Timelines, Action Items)
   - 8.3 DPO / System Admin Access Monitoring Dashboard (Anomaly Detection)
9. Deployment & Operational Architecture
   - 9.1 Target Environment Strategy (SaaS / Private Cloud)
   - 9.2 CI/CD Pipeline & Automation
   - 9.3 Observability & Monitoring (Grafana, CloudWatch, OpenTelemetry)
   - 9.4 Disaster Recovery (DR) & Backup Plan
10. Quality Assurance & Non-Functional Requirements (NFRs)
    - 10.1 Performance Metrics & SLAs
    - 10.2 Scalability & High Availability Matrix
    - 10.3 Medical Validation Strategy (Medical Acceptance Test / Expert Consensus)
    - 10.4 Safety "Fallback" Rules
11. Cost & Investment Plan
    - 11.1 Infrastructure Licensing & Cloud Costs
    - 11.2 Implementation & Professional Services

1. Executive Summary

1.1 Introduction

This technical proposal outlines the design, architecture, and implementation strategy for the **Smart Health Checkup Recommendation Dashboard** at Navavej Hospital. The primary objective is to develop an intelligent, AI-driven platform that systematically analyzes a patient's historical medical data to recommend highly accurate, safe, and personalized health checkup programs. By combining international healthcare data standards (HL7 FHIR R4) with an advanced, pluggable Artificial Intelligence architecture (Model-Switching Architecture), the proposed system will elevate Navavej Hospital's services to a true "Personalized Medicine" approach. The solution inherently prioritizes

©Copyright 2025 by Vietnam Silicon

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

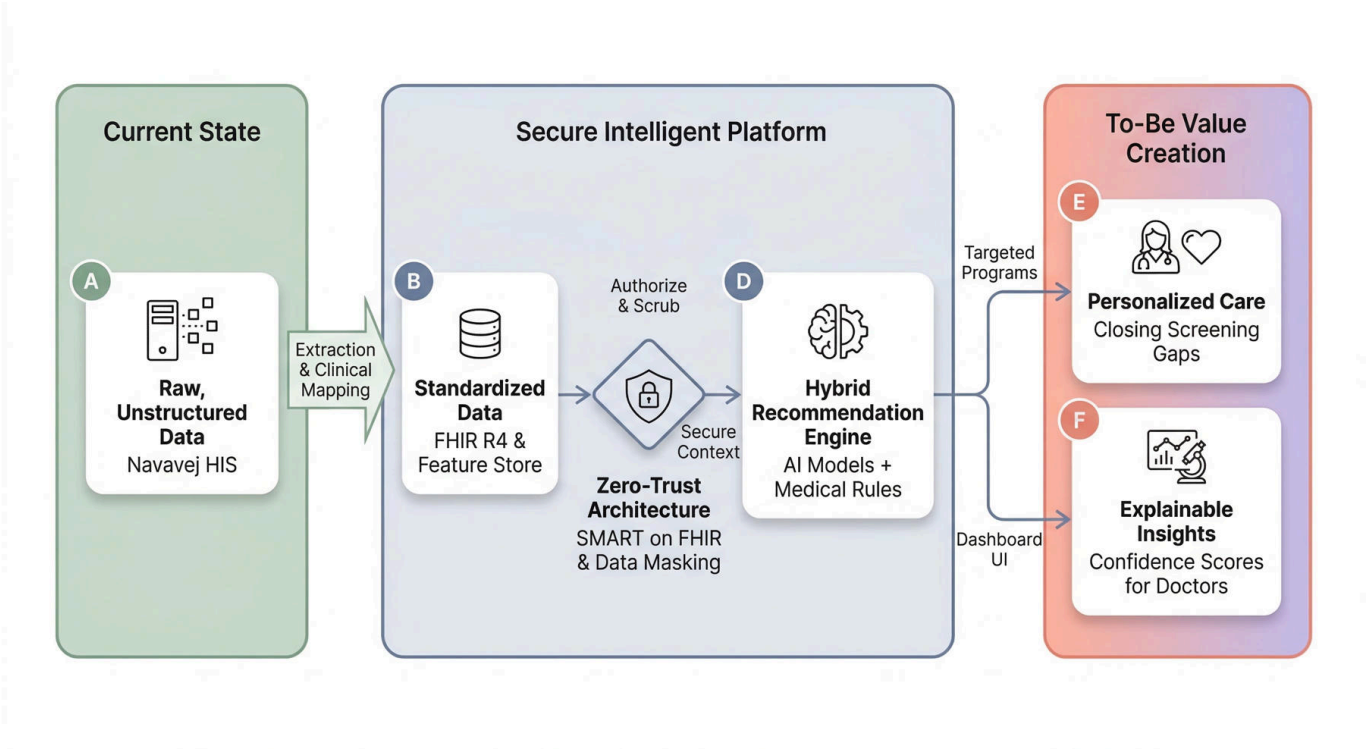
security and privacy, utilizing strict Data Masking, SMART on FHIR authorization, and the Model Context Protocol (MCP) to ensure full compliance with PDPA and HIPAA regulations.

1.2 Background Information

Traditionally, health checkup programs are selected based on generic criteria such as age and gender. However, this generalized approach frequently leads to "Screening Gaps"—missed opportunities to identify hidden health risks or prioritize critical preventative care. Service recipients possess rich, longitudinal medical histories stored within the Navavej Hospital Information System (HIS / IQVIA SaaS), encompassing diagnoses, lab value velocities, biomarker trends (e.g., eGFR slope or BMI), and historical screening data.

To bridge the gap between this wealth of raw data and actionable clinical insights, Navavej requires a system capable of interpreting complex medical histories efficiently and securely. This entails moving away from manual, ad-hoc program selection toward a **Hybrid Recommendation Engine (Rule-based combined with AI)**. The core challenge to realizing this vision is not only generating medically accurate recommendations but also standardizing the legacy data using international clinical code mapping (LOINC, SNOMED CT, ICD-10) and strictly safeguarding patient privacy before any data interacts with external AI models. This proposal presents a modular, highly secure technical blueprint designed specifically to leapfrog these challenges, while supporting seamless future AI upgrades without core system disruptions.

1.3 To-Be Vision & Core Value Proposition



The "To-Be" vision is a fully automated, transparent, and standardized workflow where raw clinical data transforms into highly personalized patient care without compromising a single data privacy boundary. The core value propositions of this platform are:

1. **Precision and Personalization:** The system will evaluate dozens of hard clinical markers (e.g., BMI velocity, JNC 8 blood pressure categories, genetic histories) to generate customized primary and "Add-On" examination recommendations that successfully close vital Screening Gaps.
2. **Explainable AI (XAI) for Trust:** Both healthcare providers and service recipients will receive clear, logical reasoning behind every AI recommendation alongside a Confidence Score, completely eliminating "black-box" medical uncertainty.
3. **Zero-Trust Security by Design:** Applying the SMART on FHIR authorization standard and the strict Least Privilege Model Context Protocol (MCP) ensures that PII is masked before any inference occurs, and every AI prompt is immutably logged for comprehensive auditing.
4. **Future-Proof Pluggable Architecture:** Exploring a Model-Switching AI Gateway guarantees Navavej Hospital flexibility. Administrators can effortlessly transition between frontier AI models (e.g., from GPT-4 to Claude 3.5 or an on-premise Local Llama model) via

©Copyright 2025 by Vietnam Silicon

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

configuration screens, entirely avoiding vendor lock-in as technology advances.

2. Project Scope & Assumptions

2.1 Scope of Work (In-Scope Deliverables)

The implementation of the Smart Health Checkup Recommendation Dashboard encompasses the complete end-to-end delivery of the data, security, AI, and presentation layers.

Implementation Layer	Deliverables & Included Scope
Data Integration & Storage	• Secure data extraction from Navavej HIS (IQVIA SaaS) via CDC sync. • Automated processing and terminology mapping into international clinical codes (LOINC, SNOMED CT, ICD-10). • Setup of the dual-storage architecture: Core **FHIR Database** & analytical **Trending Store DB**.
AI Recommendation Engine	• Development of the **Hybrid Recommendation Engine** combining hard medical rules with AI logic. • Delivery of the **AI Gateway** supporting pluggable, model-switching architecture. • Implementation of the **Model Context Protocol (MCP) Server** for secure context retrieval.
Security & Compliance	• **SMART on FHIR** authorization server acting as the gateway for all FHIR data. • Strict **Data Masking** protocol within the MCP layer to eliminate PII leakage to the AI. • Implementation of the immutable **5-Dimension Audit Log** (Who, When, Where, What, Why).
User Interfaces (Web)	• **Healthcare Provider Dashboard:** Screens displaying Individual Summaries, historical Trend Timelines (3-5 years), Screening Gaps, and Explainable AI reasoning. • **Admin/DPO Dashboard:** Screens for Real-Time Access Monitoring and Behavior Anomaly Detection.
Testing & Deployment	• Complete system containerization (Docker/Microservices). • **Medical Acceptance Testing (MAT)** to ensure the AI hits the > 95% threshold of expert consensus. • Performance testing to ensure strict SLA compliance (< 10-second latency).

2.2 Out of Scope

To ensure strict project boundaries and focus, the following items are excluded from this implementation phase:

Out-of-Scope Area	Description
Legacy HIS Replacement	The proposed solution acts as an intelligent add-on. It does not replace, refactor, or migrate the core functionality of the existing Navavej Hospital Information System (IQVIA).
Bi-Directional EMR Writes	The AI operates purely in a read-only state via the MCP server to deduce recommendations. It will not autonomously write back to or alter the core medical records in the HIS.

@Copyright 2025 by Vietnam Silicon

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

Financial / Billing AI	The AI scope is strictly limited to medical health checkup recommendations. Predictive models for hospital financial forecasting or billing are excluded.
Physical Hardware	Procurement, delivery, or maintenance of physical on-premise hardware servers is excluded (presuming standard cloud infrastructure usage).

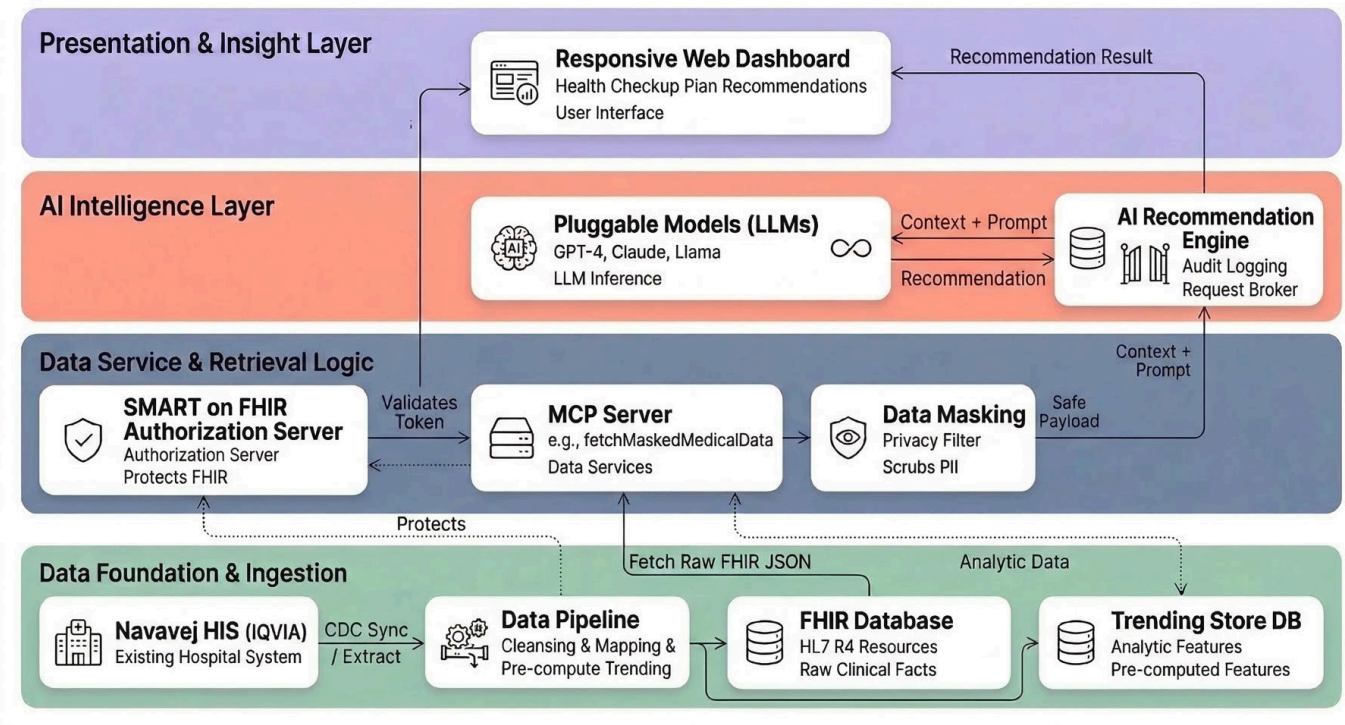
2.3 Key Assumptions & Dependencies

The successful delivery of this architecture relies on the following baseline assumptions:

Dependency Domain	Assumption Statement
Data Availability	The existing Navavej HIS (IQVIA SaaS) supports and permits reliable, near-real-time data synchronization (e.g., CDC or robust APIs) necessary to populate the pipeline.
Medical Ruleset Support	Clinical experts at Navavej will provide the definitive "Safety Rules" (e.g., age limits, gender-specific checks) required to train and constrain the rule-based safety engine.
LLM Infrastructure	Commercial LLM API accounts (such as OpenAI/Anthropic) or provisioned local GPU hardware (if utilizing a localized LLM) will be funded or provided by the hospital.

3. High-Level System Architecture

3.1 Data flow



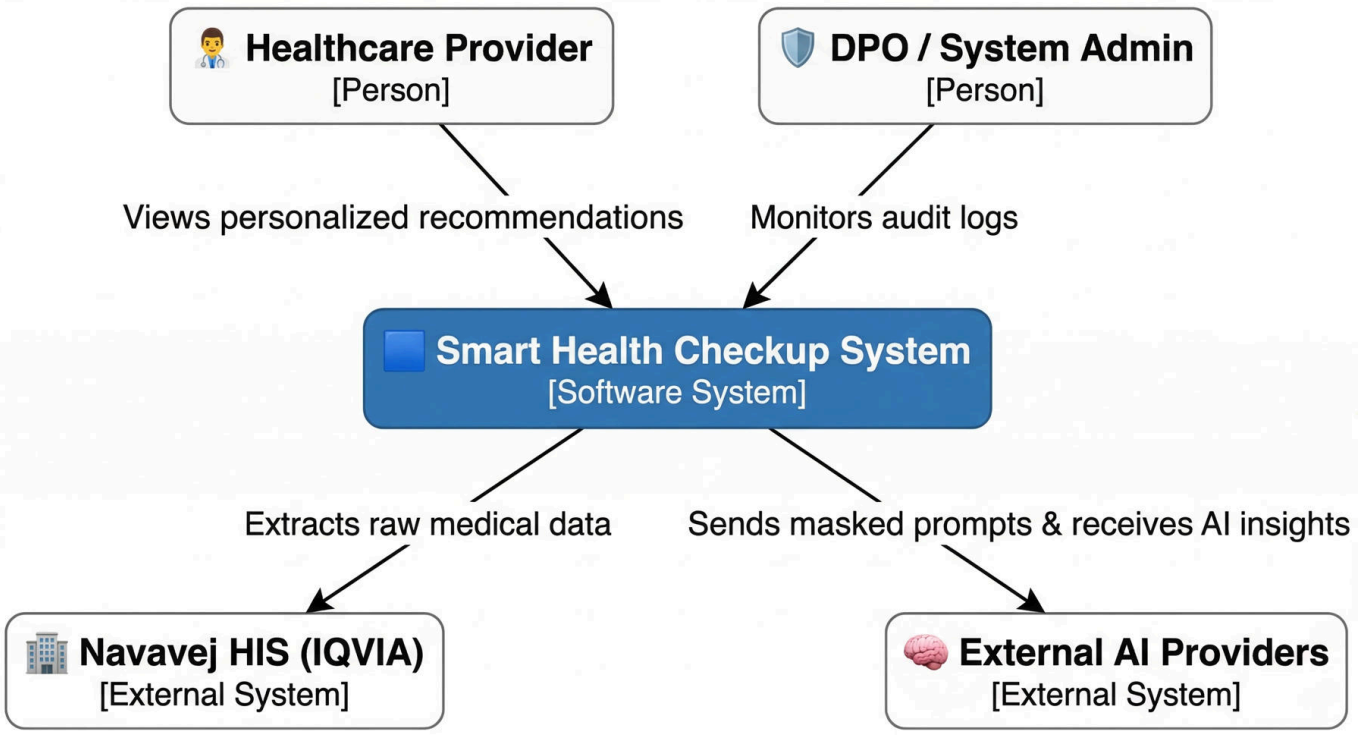
3.2 Conceptual Architecture

C4 Context Diagram

@Copyright 2025 by Vietnam Silicon.

This diagram maps out the high-level interactions between the users, the proposed system, and the external legacy systems.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon



mermaid

```
flowchart TD
    %% Nodes
    User(["🧑‍⚕️ Healthcare Provider<br>[Person]"])
    Admin(["🛡️ DPO / System Admin<br>[Person]"])

    System["🟦 Smart Health Checkup System<br>[Software System]"]

    HIS["🏢 Navavej HIS (IQVIA)<br>[External System]"]
    LLM["🧠 External AI Providers<br>[External System]"]

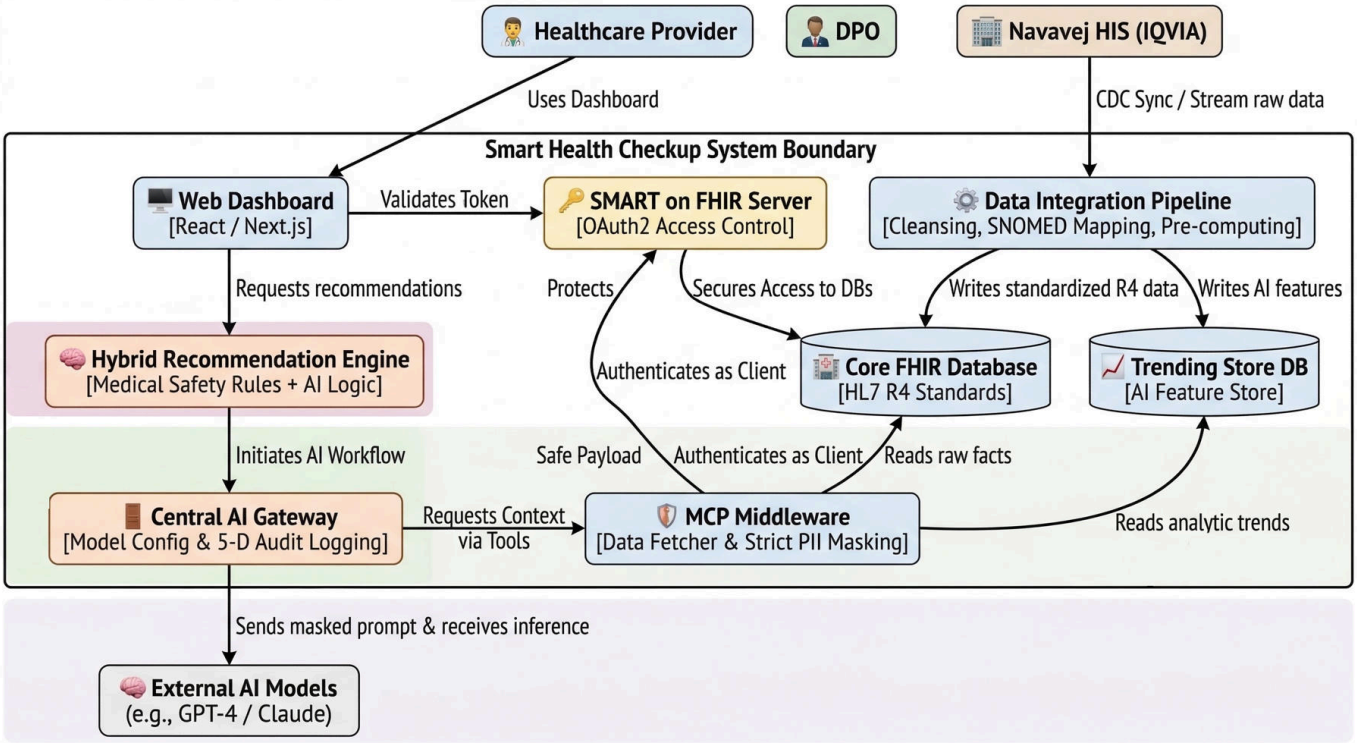
    %% Relationships
    User -->|Views personalized recommendations| System
    Admin -->|Monitors audit logs| System
    System -->|Extracts raw medical data| HIS
    System -->|Sends masked prompts & receives AI insights| LLM
```

C4 Container Diagram

This diagram zooms inside the system boundary to show the high-level technical architecture, demonstrating how the Pluggable AI, dual-databases, and MCP Middleware interact safely.

@Copyright 2025 by Vietnam Silicon.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon



mermaid

```
flowchart TD
    %% Actors and External Systems
    Provider(["👨‍⚕️ Healthcare Provider"])
    HIS["🏢 Navavej HIS (IQVIA)"]
    LLM["🧠 External AI Models (e.g., GPT-4 / Claude)"]

    %% System Boundary
    subgraph SystemBoundary [Smart Health Checkup System Boundary]
        direction TB

        WebApp["🖥️ Web Dashboard<br>[React / Next.js]"]
        Auth["🔑 SMART on FHIR Server<br>[OAuth2 Access Control]"]

        Pipeline["⚙️ Data Integration Pipeline<br>[Cleansing, SNOMED Mapping, Pre-computing]"]

        FHIR[("🏥 Core FHIR Database<br>[HL7 R4 Standards]")]
        Trend[("📈 Trending Store DB<br>[AI Feature Store]")]

        RecEngine["🧠 Hybrid Recommendation Engine<br>[Medical Safety Rules + AI Logic]"]
        AIGateway["🚪 Central AI Gateway<br>[Model Config & 5-D Audit Logging]"]
        MCP["🛡️ MCP Middleware<br>[Data Fetcher & Strict PII Masking]"]
    end

    %% Relationships
    Provider -->|Uses Dashboard| WebApp
    WebApp -->|Authenticates via| Auth
    WebApp -->|Requests recommendations| RecEngine

    HIS -->|CDC Sync / Stream raw data| Pipeline
    Pipeline -->|Writes standardized R4 data| FHIR
    Pipeline -->|Writes AI features| Trend

    Auth -.->|Secures Access to DBs| FHIR

    RecEngine -->|Initiates AI Workflow| AIGateway
    AIGateway -->|Requests Context via Tools| MCP

    MCP -->|Authenticates as Client| Auth
    MCP -->|Reads raw facts| FHIR
    MCP -->|Reads analytic trends| Trend

    AIGateway -->|Sends masked prompt & receives inference| LLM
```

@Copyright 2025 by Vietnam Silicon.

3.3 Core Modules Overview

To realize the "To-Be" architecture, the system is logically divided into 7 specialized, decoupled microservice modules:

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

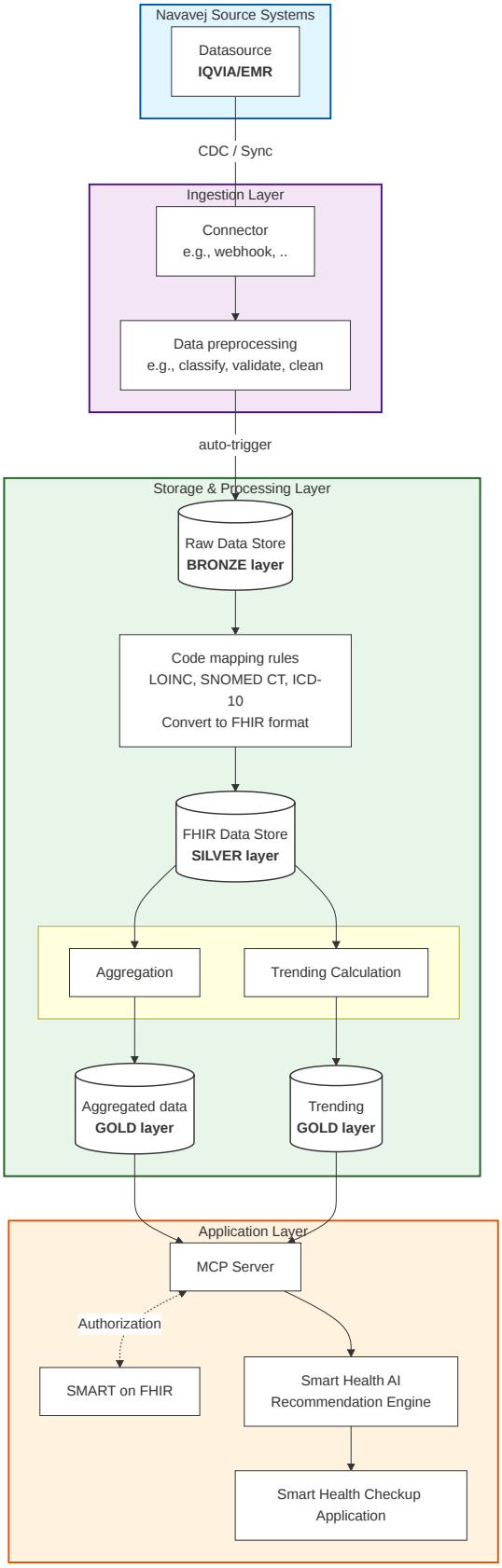
1. **Web Dashboard (Frontend UI):** A responsive application (e.g., React/Next.js) serving 2 primary user groups. It provides:
   - Sales/Doctors: with the **Individual Summary** and **Trend Timelines**
   - DPO: with an **Access Monitoring** page for anomaly detection.
2. **Data Integration Pipeline:** A robust ETL engine that captures updates from the Navavej HIS via CDC sync. It handles data cleansing, strict clinical code mapping (LOINC, SNOMED CT, ICD-10), and the pre-computation of complex health trends (e.g., eGFR slope or BMI velocity).
3. **Dual-Storage Databases:** The data storage layer is deliberately split. The **Core FHIR DB** maintains immutable, standardized clinical records, while the **Trending Store DB** acts as a flexible AI Feature Store for pre-computed longitudinal data.
4. **SMART on FHIR Authorization Server:** The centralized OAuth 2.0 gatekeeper. It ensures that every request pulling from the databases—whether requested by a doctor's dashboard or the MCP context fetcher—operates under strict, resource-level scopes.
5. **Hybrid Recommendation Engine:** The orchestrator of the system's clinical logic. It applies deterministic "Safety Rules" (e.g., specific age/gender limits) *before and after* invoking the AI, ensuring medical safety acts as a fail-safe parachute alongside the statistical AI model.
6. **Model Context Protocol (MCP) Middleware:** A specialized, Read-Only data fetcher. It serves as the ultimate privacy filter, pulling records from the FHIR and Trending databases and strictly scrubbing all Personally Identifiable Information (PII) before the context is ever shared with the AI.
7. **AI Gateway & Pluggable Models:** A model-agnostic API router that bridges the masked context to frontier LLMs (e.g., GPT-4, Claude 3.5, Local Llama). Crucially, this module is also responsible for executing the immutable 5-Dimension Audit Logging (Who, When, Where, What, Why) for every single AI inference execution.

@Copyright 2025 by Vietnam Silicon.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

4. Data Platform & Integration Architecture

High-level Data Platform Architecture




```
flowchart TD
    %% Style Definitions
    classDef boundarySource fill:#e1f5fe,stroke:#01579b,stroke-width:2px;
    classDef boundaryIngest fill:#f3e5f5,stroke:#4a148c,stroke-width:2px;
    classDef boundaryApp fill:#fff3e0,stroke:#e65100,stroke-width:2px;
```

@Copyright 2025 by Vietnam Silicon.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

```
    classDef boundaryData fill:#e8f5e9,stroke:#1b5e20,stroke-width:2px;

    classDef box fill:#ffffff,stroke:#333,stroke-width:1px,rx:0,ry:0;
    classDef db fill:#ffffff,stroke:#333,stroke-width:1.5px,rx:0,ry:0;

    %% Source System
    subgraph BoundaryDS [Navavej Source Systems]
        DS[Datasource<br/><b>IQVIA/EMR</b>]:::box
    end
    class BoundaryDS boundarySource;

    %% Ingestion Layer
    subgraph BoundaryIngest [Ingestion Layer]
        WH[Connector<br/>e.g., webhook, ..]:::box
        E[Data preprocessing<br/>e.g., classify, validate, clean]:::box
        WH --> E
    end
    class BoundaryIngest boundaryIngest;

    %% Storage & Processing Layer
    subgraph BoundaryData [Storage & Processing Layer]
        direction TB
        B[(Raw Data Store<br/><b>BRONZE layer</b>)]:::db
        M[Code mapping rules<br/>LOINC, SNOMED CT, ICD-10<br/>Convert to FHIR format]:::box
        S[(FHIR Data Store<br/><b>SILVER layer</b>)]:::db

        %% Silver rẽ nhánh cân đối sang 2 bên
        subgraph ProcessingNodes [ ]
            direction LR
            AG[Aggregation]:::box
            T[Trending Calculation]:::box
        end
        style ProcessingNodes fill:none,stroke:none

        G1[(Aggregated data<br/><b>GOLD layer</b>)]:::db
        G2[(Trending<br/><b>GOLD layer</b>)]:::db

        B --> M
        M --> S
        S --> AG
        S --> T
        AG --> G1
        T --> G2
    end
    class BoundaryData boundaryData;

    %% Application Layer
    subgraph BoundaryConsumer [Application Layer]
        direction TB
        MCP[MCP Server]:::box
        SF[SMART on FHIR]:::box
        AI[Smart Health AI<br/>Recommendation Engine]:::box
        APP[Smart Health Checkup Application]:::box

        MCP <-. Authorization .-> SF

        MCP --> AI
        AI --> APP
    end
    class BoundaryConsumer boundaryApp;

    DS ---|CDC / Sync| WH
    E -->|auto-trigger| B

    G1 --> MCP
    G2 --> MCP
```

4.1 Data Ingestion Strategy (Connecting to Navavej HIS)

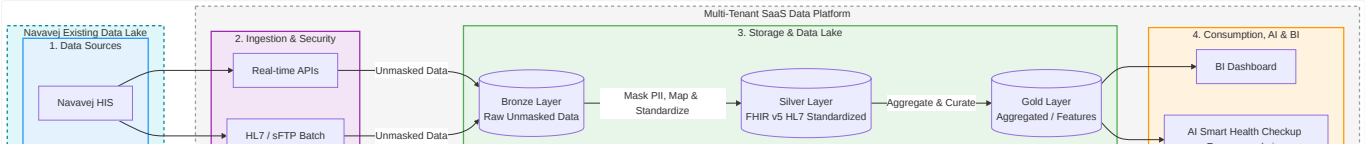
To ensure near real-time data synchronization without impacting Navavej's busy hospital operations, we implement a **Change Data Capture (CDC)** strategy.

- **CDC Extraction:** Utilizing enterprise-grade CDC tools (e.g., Debezium or AWS DMS), we stream continuous, row-level modifications directly from the IQVIA Operational Store or the HIS backend via transaction logs (log-based extraction), avoiding heavy queries on the operational databases.
- **Hybrid Ingestion:** The platform also supports Real-Time APIs for immediate checkup requests and secure Batch processing (HL7 / sFTP) for legacy data and bulk un-structured document ingestion.
- **Routing & Integrity:** The pipeline intelligently tags and routes data (differentiating new vs. existing patients) with guaranteed exactly-once or at-least-once message delivery and strict chronological ordering.

@Copyright 2025 by Vietnam Silicon.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

4.2 Data Pipeline at a Glance



mermaid

```
flowchart LR
%% Existing Infrastructure
    subgraph NavavejLake [Navavej Existing Data Lake]
        subgraph Sources [1. Data Sources]
            HIS1[Navavej HIS]
        end
    end
%% Multi-Tenant Platform
subgraph VNSPlatform [Multi-Tenant SaaS Data Platform]
    %% Ingestion Layer
    subgraph Ingestion [2. Ingestion & Security]
        API[Real-time APIs]
        Batch[HL7 / sFTP Batch]
    end

    %% Storage & Standardization Layer (Medallion Architecture)
    subgraph Storage [3. Storage & Data Lake]
        Bronze[(Bronze Layer\nRaw Unmasked Data)]
        Silver[(Silver Layer\nFHIR v5 HL7 Standardized)]
        Gold[(Gold Layer\nAggregated / Features)]

        Bronze -->|Mask PII, Map & Standardize| Silver
        Silver -->|Aggregate & Curate| Gold
    end

    %% Unified Consumption & AI Layer
    subgraph Consumption [4. Consumption, AI & BI]
        BI[BI Dashboard]
        AIRec[AI Smart Health Checkup Recommendation]
    end
end

%% Flow connections
HIS1 --> API
HIS1 --> Batch

API -->|Unmasked Data| Bronze
Batch -->|Unmasked Data| Bronze

Gold --> BI
Gold --> AIRec

%% Styling
classDef lake fill:#e0f7fa,stroke:#00838f,stroke-width:2px,stroke-dasharray: 5 5;
classDef platform fill:#f5f5f5,stroke:#9e9e9e,stroke-width:2px,stroke-dasharray: 5 5;
classDef layer1 fill:#e3f2fd,stroke:#2196f3,stroke-width:2px;
classDef layer2 fill:#f3e5f5,stroke:#9c27b0,stroke-width:2px;
classDef layer3 fill:#e8f5e9,stroke:#4caf50,stroke-width:2px;
classDef layer4 fill:#fff3e0,stroke:#ff9800,stroke-width:2px;

class NavavejLake lake
class VNSPlatform platform
class Sources layer1
class Ingestion layer2
class Storage layer3
class Consumption layer4

```

The Data Pipeline employs a modern **Medallion Lakehouse Architecture** integrated with Event-Driven processing to transform raw hospital data into AI-ready insights securely:

1. **Ingestion Layer:** Captures raw, unmasked data from Navavej HIS via APIs, Batch, and CDC, landing it directly in the **Bronze Layer**.
2. **Standardization & Masking (Bronze to Silver):** A highly secure pipeline intercepts the raw data, applies strict PII Tokenization/Dynamic Masking to comply with Thailand PDPA and HIPAA, and reformats the clinical elements into HL7 FHIR R4 standards before storing it in the **Silver Layer**.
3. **Curation & Aggregation (Silver to Gold):** Automated ETL/ELT workflows transform the standardized FHIR data into pre-aggregated health metrics, time-series trends, and ML features within the **Gold Layer**, optimized for BI and AI consumption.

@Copyright 2025 by Vietnam Silicon.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

4.3 Data Conversion & Health Trend Pre-Computing

4.3.1 Clinical Code Mapping (LOINC, SNOMED CT, ICD-10)

A dedicated service converts "Local Codes" from Navavej's HIS into international standard codes:

- **LOINC:** For laboratory test result codes (Observations).
- **SNOMED CT:** For symptoms, diagnoses, and procedures (Conditions/Procedures).
- **ICD-10:** For disease classification.

Technical Implementation:
We deploy a robust **FHIR Terminology Server** (e.g., HAPI FHIR Terminology, Ontoserver) within the pipeline. This service automatically ingests FHIR `ConceptMap` and `ValueSet` resources to translate Navavej's proprietary HIS codes and Thai national standard codes (TMT, TMLT, TCHI) dynamically into the globally recognized standards during the ingestion phase with sub-second latency.

4.3.2 Health Trend Pre-Computing (Velocity, Gaps, eGFR slope)

To optimize AI performance and ensure reliable medical accuracy, longitudinal calculations (e.g., BMI velocity over 12 months, eGFR slope decline) are explicitly pre-computed during this pipeline phase. Rather than relying on the AI to deduce complex math on the fly, these generated features are stored in a secondary, specialized **Trending Store DB** (an AI Feature Store). This enterprise pattern keeps the primary FHIR Database beautifully pristine for immutable clinical records while the Trending Store securely holds analytical AI features ready for instant AI context mapping.

Technical Implementation:
We construct an automated ETL/ELT pipeline leveraging distributed compute frameworks (e.g., Apache Spark or dbt). This architecture utilizes event-driven triggers; immediately upon the arrival of new clinical data via CDC, micro-batch processes spin up to recalculate critical health metrics and sync the updated features to the Gold Layer.

4.4 Dual-Storage Strategy

4.4.1 Core FHIR Database (Immutable Clinical Records)

The Core FHIR Database represents our **Silver Layer**, serving as the system of record for standardized, masked clinical data.

- **Design Pattern:** Data is strictly modeled as HL7 FHIR R4 JSON resources.
- **Immutability & Compliance:** All Personally Identifiable Information (PII) is removed or tokenized prior to insertion. The database is treated as an append-only, immutable ledger ensuring auditability, exact historical clinical states, and uncompromising adherence to PDPA regulations.

4.4.2 Trending Store Database (AI Analytical Feature Store)

The Trending Store acts as the **Gold Layer** and serves as a high-performance **AI Feature Store**.

- **Design Pattern:** Optimized for heavy read-access, time-series querying, and columnar analytics rather than transactional clinical updates.
- **Purpose:** It holds the pre-aggregated metrics, statistical gaps, and engineered features (like health velocity and moving averages). This allows the application layer, BI Dashboards, and AI models to retrieve rich patient context with millisecond latency without querying the Core FHIR Database.

5. Data Platform Tech Stack

Our solution is flexible and can be deployed utilizing either AWS Managed Services or an Open-Source Hybrid approach:

AWS Cloud Strategy:

- **Ingestion & CDC:** AWS DMS (Database Migration Service) or Managed Kafka (MSK).
- **Data Storage (Bronze/Silver/Gold):** Amazon S3 (Bronze), AWS HealthLake (Native FHIR Silver Layer), and Amazon Redshift / Athena (Gold/Trending Store).
- **Processing & PII Masking:** AWS Lambda for dynamic masking, AWS Glue / EMR (Apache Spark) for ETL and trend pre-computation.
- **Serving & Security:** Amazon API Gateway, Amazon Cognito (OAuth 2.0 / RBAC), AWS KMS for encryption.

Open Source / Hybrid Strategy:

@Copyright 2025 by Vietnam Silicon.

- **Ingestion & CDC:** Debezium and Apache Kafka.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

- **Data Storage (Bronze/Silver/Gold):** MinIO (S3-compatible Bronze), HAPI FHIR backed by PostgreSQL (Silver Layer), and ClickHouse (High-performance Gold/Trending Store).
- **Processing & Orchestration:** Apache Airflow / Temporal for orchestration, Apache Flink for real-time masking, and dbt for metric aggregations.
- **Serving & Security:** Kong / KrakenD API Gateway, Keycloak for Identity/OAuth, and HashiCorp Vault for secure vaults and re-identification.

6. Security, Privacy & Data Governance

6.1 SMART on FHIR Authorization (Access Control Gatekeeper)

TBD *Diep*

6.2 Data Masking & Pseudo-Anonymization Pipelines

TBD *Diep + Data*

6.3 Comprehensive Audit Logging

TBD *Diep + Nguyen*

7. AI & Recommendation Engine Architecture

7.1 High-Level AI Workflow

TBD *Nguyen*

7.2 Model Context Protocol (MCP) Middleware (Secure Context Retrieval)

TBD *Nguyen*

7.3 Hybrid Recommendation Logic (Rule-based Safety + AI Inference)

TBD *Nguyen*

7.4 Central AI Gateway

TBD *Nguyen*

7.4.2 Pluggable LLM Configuration Management

TBD *Nguyen*

7.4.1 Explainable AI (XAI) & Confidence Scoring Generation

TBD *Nguyen*

7.5 AI Tech Stack

TBD *Nguyen*

8. Frontend & Dashboards (User Experience)

8.1 Responsive Web Application Architecture (Tech Stack)

TBD *Diep*

8.2 Healthcare Provider Dashboard (Summaries, Timelines, Action Items)

TBD *Diep*

8.3 DPO / System Admin Access Monitoring Dashboard (Anomaly Detection)

TBD *Diep*

@Copyright 2025 by Vietnam Silicon.

This document is confidential and restricted to Vietnam Silicon use only; it cannot be shared with third parties without prior written authorization from Vietnam Silicon

9. Deployment & Operational Architecture

TBD *Diep*

9.1 Target Environment Strategy (SaaS / Private Cloud)

9.2 CI/CD Pipeline & Automation

9.3 Observability & Monitoring (Grafana, CloudWatch, OpenTelemetry)

9.4 Disaster Recovery (DR) & Backup Plan

10. Quality Assurance & Non-Functional Requirements (NFRs)

TBD *Diep*

10.1 Performance Metrics & SLAs

10.2 Scalability & High Availability Matrix

10.3 Medical Validation Strategy (Medical Acceptance Test / Expert Consensus)

10.4 Safety "Fallback" Rules

11. Cost & Investment Plan

TBD *Diep*

11.1 Infrastructure Licensing & Cloud Costs

11.2 Implementation & Professional Services

@Copyright 2025 by Vietnam Silicon.